INFO-562 AI-Driven Decision Support Sys
Assignment 3 (100 points)

Deadline: April 10, 2025 11:59 pm in Brightspace

Please remember to include the following statement at the beginning of your submitted assignment and SIGN it. Your assignment won't be graded without the signed statement.
"I have done this assignment completely on my own. I have not copied it, nor have I given my solution to anyone else. I understand that if I am involved in plagiarism or cheating, I will have to sign an official form that I have cheated and that this form will be stored in my official university record. I also understand that I will receive a grade of 0 for the involved assignment and my grade will be reduced by one level (e.g., from A to A- or from B+ to B) for my first offence, and that I will receive a grade of "F" for the course for any additional offence of any kind."

Assignment 3 is an extension of the assignment 2. You can refer to assignment 2 for definition and further clarifications. **Download Data file from the assignment 2.**

Data Description: The provided **dataset** reports the weekly sales of a tech-gadget e-commerce retailer over a period of 98 weeks, from October 2016 to September 2018. It includes the weekly sales of 44 items, also called stock keeping units (SKUs), as well as diverse information on these SKUs. The dataset is **preprocessed** and ready to use for computing such as predictions, etc.

There are advantages and disadvantages when aggregating data across SKUs. Aggregating sales data across all SKUs reduces the noise and allows the model to rely on a larger number of observations but will overlook the fact that different SKUs have differing characteristics. There is also a clear trade-off between performance and running time. Based on this observation, one may want to consider a compromise between the two extreme approaches (centralized and decentralized) by aggregating a group of similar SKUs together. A natural way to do so is by using clustering techniques. Here you will be implementing two common clustering techniques: (1) **k-means** and (2) **density-based spatial clustering of applications with noise** (**DBSCAN**).

Q. 1: (45 Points) You will be implementing k-means clustering.

First you need to create groups of SKUs that share similar characteristics and will be assigned to the same cluster. To this end, you will apply the k-means clustering methods on the 44 SKUs in the provided dataset. You should divide the SKUs into k clusters and subsequently estimate a demand prediction model (e.g., OLS - ordinary least squares) for each cluster.

a. Algorithm should run using the average values of the price and weekly sales of each SKU: first create a table where the rows are the different SKUs, and the columns are the average values of the price and weekly sales variables. Scaling of the different features has a significant impact on the distance function (which is used to find the closest center), You should scale the values Min Max scaler (specifically, use the MinMaxScaler from sklearn). You should follow the example table below to arrange this data for SKU 1 and the price and weekly sales features.

	Price (scaled)	Weekly sales (scaled)
SKU 1	Scaled average price of SKU 1 over the training period	Scaled average weekly sales of SKU 1 over the training period
...		

You should apply the k-means clustering method on this table to aggregate similar SKUs. Following this process, each SKU is assigned to one specific cluster (the clusters identified by k-means are non-overlapping by design). The parameter k is a parameter (or user defined/selected input) of the model.

b. Estimate a centralized OLS for each cluster: For each cluster, you will estimate an OLS regression (using the observations from all the SKUs assigned to the cluster). This is equivalent to assuming that all the SKUs in the same cluster have the same demand prediction model.

c. You Python runs for Clustering using Average Price and Weekly Sales and print the following results.

Best Model:
Number of clusters:
Validation R^2:
OOS R^2:

d. Now include both the average and standard deviation of the price and weekly sales into the clustering method, most of the code is the same as before, except that the clustering step needs to be slightly adapted to accommodate average and standard deviation. Print the improve results after adding the standard deviation of the clustering features (i.e., price and weekly sales) as follows:

Best Model:
Number of clusters:
Validation R^2:
OOS R^2:
Time to compute:

e. Plot (e.g., scatterplot) weekly sales as a function of the price.

Q. 2: (45 Points) You will be implementing **density-based spatial clustering of applications with noise** (**DBSCAN**) clustering. You should review literature to learn about the DBSCAN.

a. You Python code runs for DBSCAN Clustering using Average Price and Weekly Sales and print the following results.

Best Model:
Parameters: [0.2, 3]
Validation R2:
OOS R2:

Time to compute:

b. Plot (e.g., scatterplot) DBSCAN clusters using above parameters.

c. Now include average and standard deviation of sales price and print the improve results after adding the standard deviation of the clustering features as in the table below:

Summary of the results for the DBSCAN clustering method

Features	Best model	OOS R^2	Computing time (sec)

Q. 3: (10 Points) Compare in tabular form and discuss merits of the results of the both the clustering algorithm.

Note: Submit Python code in .IPYNB format and PDF file with discussion on results.